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High Performance Processors and Systems

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Name
Last Name

Problem 1 (5 points)	
Problem 2 (5 points)	
Problem 3 (6 points)	
Question 1 (5 points)	
Question 2 (5 points)	
Question 3 (6 points)	
Total (32 points)	

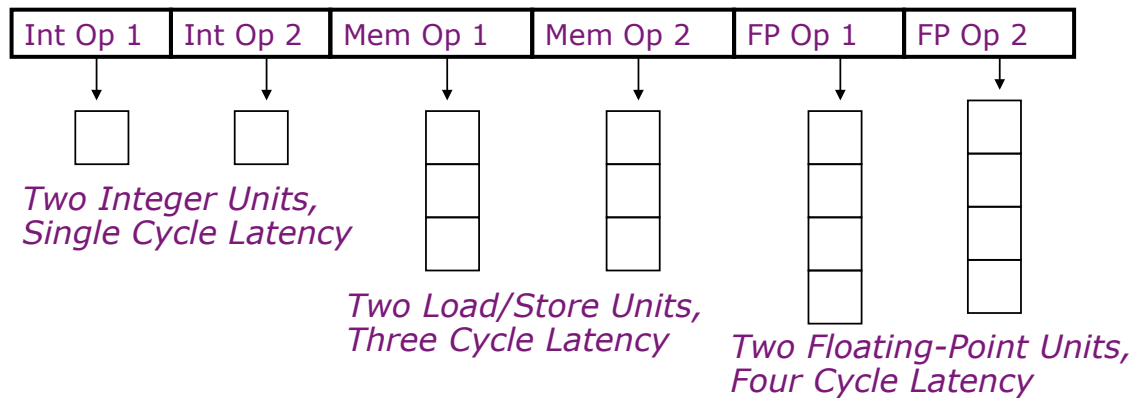
To pass the exam (the following conditions are in AND):

- The overall score has to be equal or higher than 18
- At least 1 problem has to be correct (at least it has to be evaluated with 85% of the "available" points)
- At least 9 points from the problems
- At least 9 points from the questions

Problem 1

P1.A: Explain the main differences between a VLIW architecture and a superscalar architecture, in terms of instructions scheduling, hardware complexity, and code size.

P1.B: Considering the following VLIW "architecture":



Considering the following portion of assembly, describe the corresponding VLIW code:

```
1. Loop:      LD F0, 0(R1)
               FADD F4, F0, F2
               SD F4, 0(R1)
               ADD R1, 8
               BNE R1 R2 Loop
```

How many FLOPS/cycle?

2. With an unroll equal to 4 we can obtain the following code

```
Loop:      LD F0, 0(R1)
            LD F6, 8(R1)
            LD F10, 16(R1)
            LD F14, 24(R1)
            FADD F4, F0, F2
            FADD F8, F6, F2
            FADD F12, F10, F2
            FADD F16, F14, F2
            SD F4, 0(R1)
            SD F8, 8(R1)
            ADD R1, 32
            SD F12, -16(R1)
            BNE R1 R2 Loop
            SD F16, -8(R1)
```

How many FLOPS/cycle?

Answer P1.A

Answer P1.B.1

Int1

Int 2

M1

M2

FP+

FPx

[illegible]

Answer P1.B.2

Int1

Int 2

M1

M2

FP+

FPx

[illegible]

Problem 2

Design and describe (all the decisions have to be motivated) a 1-BHT and a 2-BHT able to execute the following assembly code (R0 is set to 1000, R1 is set to 0)

```
LOOP:    LD      F1  0    R0
          ADDD    F2  F1   F1
          ADDI    R1  R1   1000
LOOP2:    MULTD   F2  F2   F1
          SUBI    R1  R1    1
          BNEZ    R1  LOOP2
          SUBI    R0  R0    1
          BNEZ    R0  LOOP
```

The obtained result, in terms of miss predictions, is inline with theoretical characteristics of the two predictors? Please effectively support your answer.

NOTE: When *making your point*, you have to use the same BHT for both the loops

Answer

Problem 3

Assume that the following code is executed on a CPU with TOMASULO and with the following units:

- 1 LOAD/STORE unit with Latency equal to 1
- 2 MULT units with latency equal to 10
- 3 ADD/SUBD unit with latency equal to 2

And the following information:

First load 8 clock cycles (cache miss)

Following loads 1 clock cycle (hit)

$$R1 = 10$$

Show the computation steps of the following lines of code:

```
Loop:    LD F0 0 R1
         ADDD F6 F4 F0
         MULTD F10 F0 F4
         ADDD F4 F10 F2
         SUBI R1 R1 #10
         BNEZ R1 Loop
```

[illegible]

Tables

[illegible][illegible][illegible]

[illegible][illegible][illegible]

t =				R1 =				
Name	Op	Vj	Vk	Qj	Qk	Etime		Busy
mult1							Load	
mult2								
add1								
add2								
add3								
F0	F2	F4	F6	F10	F12			

[illegible]

t =				R1 =						
Name	Op	Vj	Vk	Qj	Qk	Etime		Busy	Addr	
mult1							Load			
mult2										
add1										
add2										
add3										
F0	F2	F4	F6	F10	F12					

[illegible][illegible]

Name	Op	Vj	Vk	Qj	Qk	Etime
mult1						
mult2						
add1						
add2						
add3						

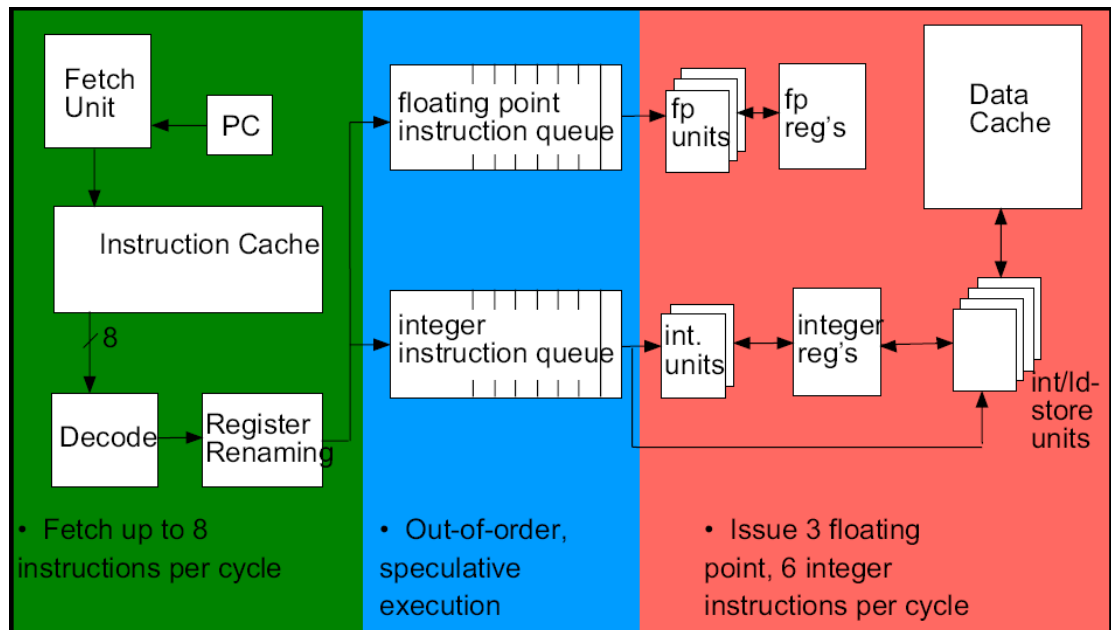
[illegible]

Name	Op	Vj	Vk	Qj	Qk	Etime
mult1						
mult2						
add1						
add2						
add3						

t =				R1 =						
Name	Op	Vj	Vk	Qj	Qk	Etime			Busy	Addr
mult1								Load		
mult2										
add1										
add2										
add3										
F0	F2	F4	F6	F10	F12					

Question 1

1. Given a processor oriented to Instruction Level Parallelism such as the one in figure, can it be exploited for thread level parallelism? Motivate your answer.
2. Which kind of Thread level parallelism is best suited and which modifications are necessary to this architecture?
1. How this can improve performance? Explain which term of the performance formula can be improved.



Question 2

In MIMD machines specify the type of memory organization available and how to choose among them. Specify then the possible memory address space models available, their advantages and disadvantages and their relation to the memory organization, if any.

Question 3

Present the different types of dependences among instructions that can occur and relate them to the possible hazards that may occur in a pipeline.

